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One-Pot Synthesis of 1,2,3-Triazolo Polycyclic Systems via Copper-Catalyzed/TsOH-Promoted Tandem Annulation of 1,6-Allenynes with Organic Azides

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Abstract: A copper-catalyzed/TsOH-promoted one-pot tandem annulation of 1,6-allenynes with organic azides was reported, providing *N*-allyl 1,2,3-triazole-fused structures in moderate to excellent yields under mild reaction conditions. Moreover, the late-stage diversification of the *N*-allyl products can be realized by chemoselectively reducing to the *N*-alkyl counterparts.

Keywords: CuAAC; one-pot protocol; tandem reaction; 1,2,3-triazole-fused structure

1,2,3-Triazole and its fused systems have gained significant attention due to their broad applications in medicinal chemistry[1] and materials science.[2] Among them, fused 1,2,3-triazole and azine ring systems as a useful class of hybrid heterocycles have interesting pharmacological activities (Figure 1).[3] Developing efficient and versatile routes toward 1,2,3-triazoloazine structures is therefore highly desirable.[4]

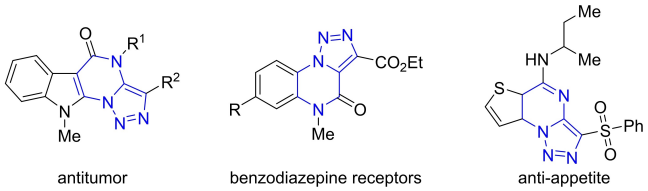
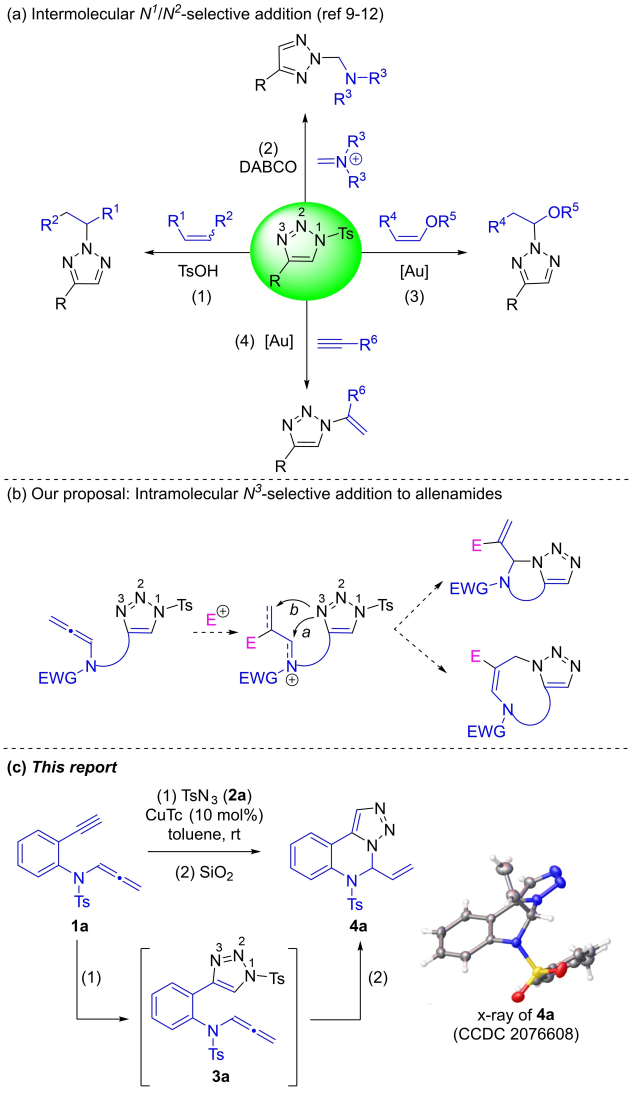


Figure 1. Examples of Biological 1,2,3-Triazoloazine Structures.

N-sulfonyl-1,2,3-triazoles, which can be easily prepared from copper-catalyzed azide-alkyne cycloadditions (CuAAC),[5] have received much attention because of their useful transformations through *α*-imino metal carbenes[6] or *N*-sulfonly ketenimines[7] concomitant with denitrogenative ring opening. However, in comparison with its *NH*-1,2,3-triazole analogues for the synthesis of *N*-substituted 1,2,3-triazoles,[8] *N*-sulfonyl-1,2,3-triazoles have been relatively less developed.[9–12] In 2016, Chen's group reported an efficient Brønsted acid-promoted N^2-selective addition of *N*-sulfonyl-1,2,3-triazoles to olefins for the synthesis of *N*-alkyl-1,2,3-triazoles (Scheme 1a, equiv. 1);[9] meanwhile, Shi's group developed a novel base-induced synthesis of *N*-dialkylaminomethyl-1,2,3-triazoles through N^2-selective addition of *N*-sulfonyl-1,2,3-triazoles to iminiums (Scheme 1a, equiv. 2);[10] later, a gold-catalyzed N^2-selective addition of *N*-sulfonyl-1,2,3-triazoles to vinyl ethers was developed by Zhao and coworkers (Scheme 1a, equiv. 3);[11] subsequently, the same group further realized a gold-catalyzed N^1-selective addition of *N*-sulfonyl-1,2,3-triazoles to alkynes, providing *N*-vinyl-1,2,3-trizoles (Scheme 1a, equiv. 4).[12] Despite these intermolecular advances, little attention has been paid to the intramolecular versions of *N*-sulfonyl-1,2,3-triazoles for the synthesis of 1,2,3-triazole-fused structures.[13]

On the other hand, *N*-allenamides are versatile building blocks and highly susceptible towards

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Scheme 1. Previous Reports and Our Proposal.

Brønsted acid or metal promoted electrophilic activation.[14] Thus, we envisioned that if the R group of *N*-sulfonyl-1,2,3-triazoles is an allenamide, the intramolecular N^3-selective nucleophilic cyclization might occur to provide *N*-allyl-1,2,3-triazole-fused structures (Scheme 1b).

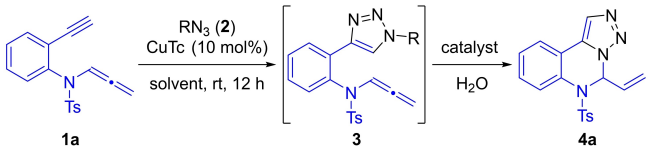
Based on the hypothesis and in continuation with our interest in allene chemistry for developing novel methodologies on polycyclic skeleton synthesis,[15] we initially tried to synthesize the allenamide-tethered *N*-sulfonyl-1,2,3-triazole **3a** via the CuAAC reaction of 1,6-allenyne **1a** with tosyl azide **2a**. Surprisingly, a novel tricyclic 1,2,3-triazoloquinazoline **4a** was regioselectively obtained as single product after silica gel chromatography (Scheme 1c). The structure of **4a** was unambiguously confirmed by X-ray diffraction analysis.[16] Herein, we report our preliminary results.

The allenamide-tethered *N*-sulfonyl-1,2,3-triazoles **3** were prepared in situ via the well-developed CuTc-catalyzed reaction of 1,6-allenyne **1a** with the corresponding organic azides **2**.[5] According to the previous reports,[9–12] trace amount of water is auxiliary to capture the leaving *p*-toluenesulfonyl (Ts) group. We first conducted the reaction in toluene at room temperature with SiO_2 as catalyst, water as addtive and obtained the desired product **4a** in 68% yield (Table 1, Entry 1). Control experiments without catalyst or with 4 Å molecular sevies (MS) as additve yielded no product (Entries 2–3). Besides *p*-toluenesulfonyl (R = Ts) group, other sulfonyl groups were also tested, while methanesulfonyl (R = Ms) group gave inferior result (Entry 4) and *p*-nitrobenzenesulfonyl (R = Ns) group afforded no product at all (Entry 5). Then, different solvents were screened. In general, aromatic solvents were better than others for this one-pot reaction (Entries 6–9). Both benzene and xylenes gave comparable results, whereas other solvents such as acetonitrile (MeCN) and tetrahydrofuran (THF) were not suitable for the reaction. At the stage of substrate investigation, however, we found that SiO_2 showed poor catalytic performance for functionalized substrates (Table 2). Thus, various acids were examined. The reaction did not work with H_2SO_4 as catalyst (Entry 10). Gratifyingly, MsOH and TsOH gave product **4a** in 55% and 69% yields respectively within 5 min (Entries 11–12). In addition, Lewis acid such as $AlCl_3$ was also tested, affording only moderate result (Entry 13). Though TsOH-promoted autocatalytic transformation of *N*-tosyl-1,2,3-triazoles was reported,[17] a decrease of the catalyst loading gave product **4a** in 47% yield (Entry 14). A longer reaction time led to no significant improvement (Entry 15), while high temperature resulted no desired product because the allenyne substrate is unstable at high temperature (Entry 16).[18]

With the optimal reaction conditions in hand, we next set out to investigate the substrate scope. Various aniline-derived 1,6-allenynes were subjected to this reaction (Table 2). First, the 1,6-allenyne substrates with other sulfonyl groups than *p*-toluenesulfonyl substituent all proceeded well to give the corresponding products in moderate to excellent yields, except that the methylsulfonyl substrate afforded the desired product **4f** in only 22% yield (**4a–f**). Ester group (R = CO_2Me) instead was also suitable for the reaction and the corresponding product **4g** was obtained in 41% yield. Then, the reaction of tosyl azide **2a** with different substituted 1,6-allenynes bearing electron-donating or electron-withdrawing substituents at the random position of the aryl moiety all proceeded smoothly to provide the desired products in moderate to excellent yields (**4h–u**). Furthermore, both di-substituted 1,6-allenynes and the bulky naphthyl one were all well tolerated for the reaction (**4v–y**). When

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Table 1. Optimization of Reaction Conditions.[a]



Entry	R	Catalyst	Solvent	Yield (%)[b]
1	Ts (**2a**)	SiO_2	toluene	68
2	Ts (**2a**)	–	toluene	–
3[c]	Ts (**2a**)	SiO_2	toluene	–
4	Ms (**2b**)	SiO_2	toluene	42
5	Ns (**2c**)	SiO_2	toluene	–
6	Ts (**2a**)	SiO_2	benzene	56
7	Ts (**2a**)	SiO_2	xylenes	61
8	Ts (**2a**)	SiO_2	MeCN	–
9	Ts (**2a**)	SiO_2	THF	–
10[d]	Ts (**2a**)	H_2SO_4	toluene	–
11[d]	Ts (**2a**)	MsOH	toluene	55
12[d]	Ts (**2a**)	TsOH	toluene	69
13[d]	Ts (**2a**)	$AlCl_3$	toluene	39
14[e]	Ts (**2a**)	TsOH	toluene	47
15[f]	Ts (**2a**)	TsOH	toluene	71
16[g]	Ts (**2a**)	TsOH	toluene	0

[a] Reaction conditions: **1a** (0.1 mmol), **2** (0.1 mmol), CuTc (10 mol%), solvent (1.0 mL), 12 h, rt; then catalyst (SiO_2, 200 mg), additive (H_2O, 0.1 mmol), 24 h.
[b] Isolated yields.
[c] 4 Å MS (100 mg) was added.
[d] Catalyst (100 mol%), 5 min.
[e] Catalyst (20 mol%), 6 h.
[f] 12 h.
[g] At 100 °C.

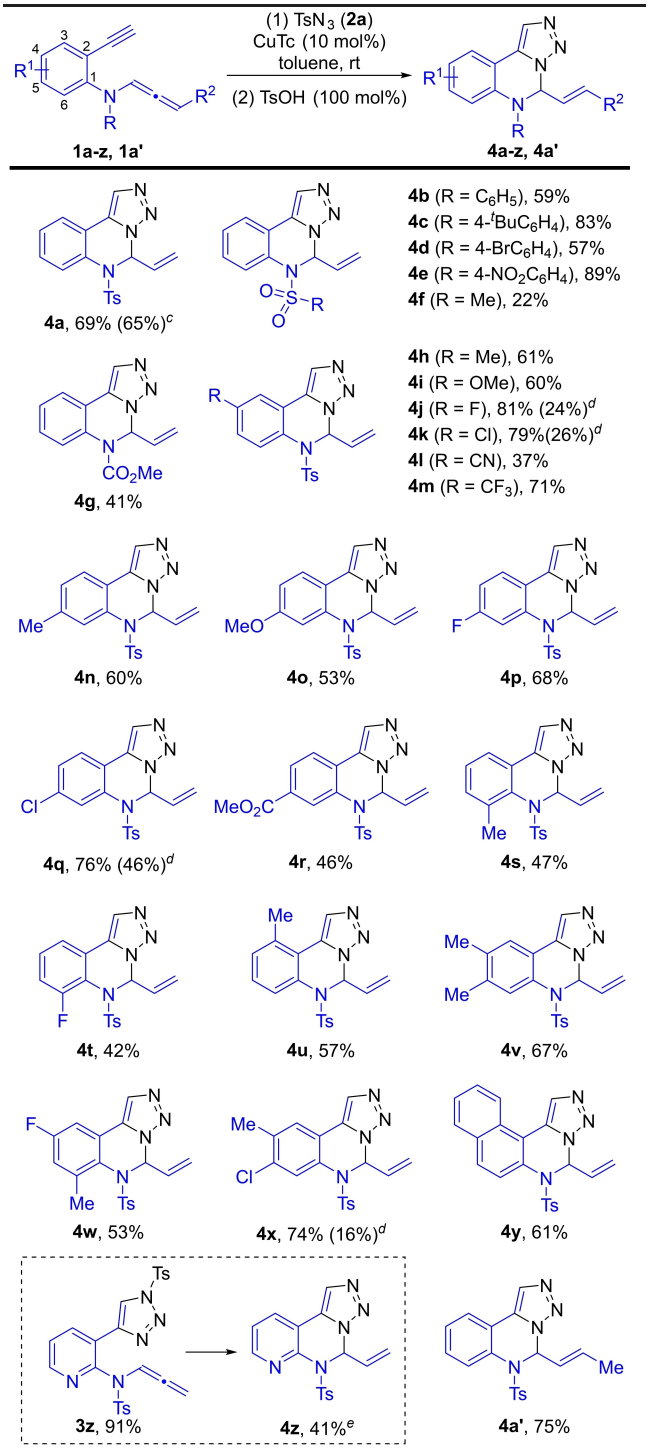
pyridine-derived 1,6-allenyne was subjected to the standard reaction conditions, the corresponding allenamide-tethered *N*-sulfonyl-1,2,3-triazole **3z** was obtained in almost quantitative yield, which could transform to **4z** in 41% yield at elevated temperature.[18] Gratifyingly, 1,6-allenyne **1a'** (R^2 = Me) furnished the desired product **4a'** in 75% yield. It is worth noting that TsOH showed better functional group compatibility than SiO_2 in most cases (**4j–k**, **4q**, and **4x**). Gram-scale synthesis was also performed under standard reaction conditions and gave product **4a** in 65% yield.

Besides the 1,6-allenyne substrates bearing an aromatic linker, benzo-free aliphatic 1,6-allenyne **1b'** was also investigated (Scheme 2a). Although the allenamide-tethered *N*-sulfonyl-1,2,3-triazole **3b'** could be isolated in 90% yield, no desired product but rather complex mixture was obtained when **3b'** was subjected to TsOH in toluene at room temperature.[19] This failure indicates that the aromatic linker is essential for the reaction and may stabilize the reaction intermediate. Gratifyingly, phenoxyallene-based 1,6-allenyne **1c'** was suitable for this reaction (Scheme 2b). The phenoxyallene-tethered *N*-sulfonyl-1,2,3-triazole **3c'** was isolated in 93% yield, further affording the desired product **4c'** in only 25% yield maybe due to the relatively poor electron-donating property of oxygen.[18] To further expand the reaction scope, we turned our attention into other two types of allenyne substrates bearing allenoate (Scheme 2c) and unbiased allene (Scheme 2d). Unfortunately, the corresponding desired products were not detected althouth the relevant *N*-sulfonyl-1,2,3-triazoles were obtained.

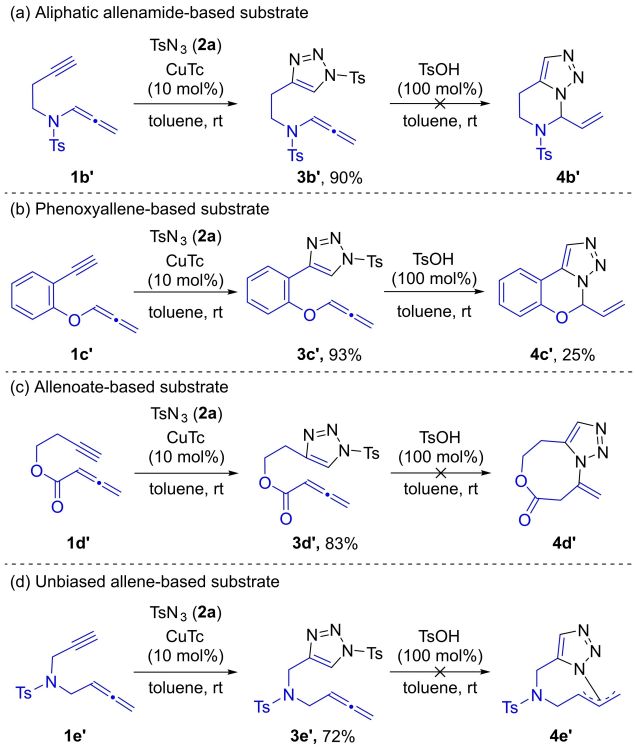
On the basis of previous reports,[9–12] two plausible catalytic cycles are depicted in Scheme 3. The allenamide-tethered *N*-sulfonyl-1,2,3-triazole **3a** is generated in situ from **1a** and **2a** via CuAAC reaction. Then, an iminium cation intermediate **I** is generated from the allenamide with the addition of a proton,[14] which is then attacked by the internal N^3-nitrogen of the triazole to give the intermediate **II**. Then, the activated S–N bond is hydrolyzed to form the desired product **4a** concomitant with the release of TsOH and a proton (path a, Scheme 3). On the other hand, the TsOH-catalyzed hydrolyzation of *N*-sulfonyl-1,2,3-triazole **3a** is possible and affords *NH*-1,2,3-triazole **III**. Then, an iminium cation intermediate **IV** is generated from the allenamide with the addition of a proton, which is then attacked by the internal N^3-nitrogen of triazole to give

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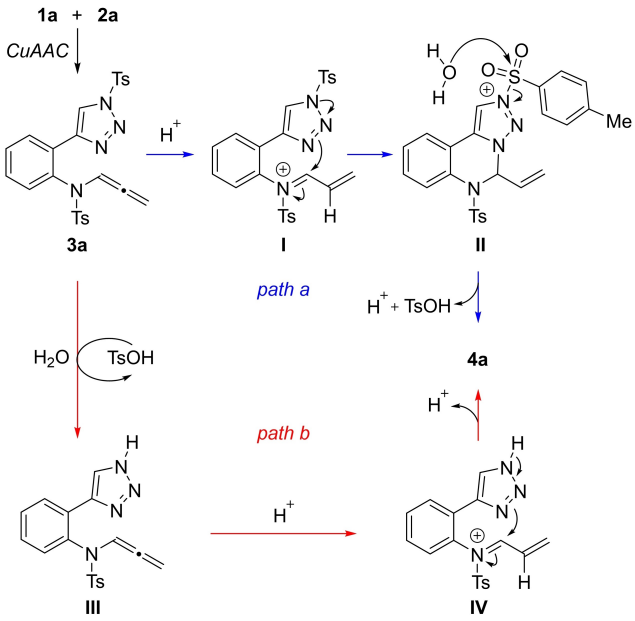
Table 2. Scope of 1,6-Allenyne Substrates.[a,b]



[a] Reaction conditions: **1** (0.2 mmol), **2a** (0.2 mmol), CuTc (10 mol%), toluene (2.0 mL), 12 h, rt; then TsOH (100 mol%), 5 min–48 h, rt.
[b] Isolated yields.
[c] 5 mmol scale.
[d] Isolated yields with SiO_2 (200 mg) as catalyst, 24 h.
[e] At 100 °C.



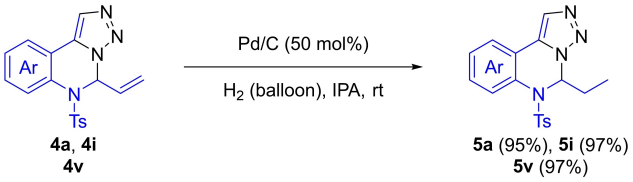
Scheme 2. Scope of Other Type of Allenyne Substrates.



Scheme 3. Possible Reaction Mechanisms.

the desired product **4a** with the release of a proton (path b, Scheme 3). Several control experiments including deuterium-labeling experiment and in-situ NMR investigation demonstrate that the concerted mechanism is more reasonable (Supporting Information).

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Scheme 4. Transformation of Products.

The versatility of this TsOH-promoted CuAAC annulation can be further exploited in chemoselective reduction (Scheme 4). The *N*-allyl-1,2,3-triazole products **4** could be readily reduced to the corresponding *N*-alkyl counterparts via Pd/C-catalyzed chemoselective hydrogenation under H_2 in isopropyl alcohol (IPA) at room temeprature.

In summary, we have developed a copper-catalyzed/TsOH-promoted one-pot tandem annulation of 1,6-allenynes with organic azides to provide *N*-allyl 1,2,3-triazole-fused structures in moderate to excellent yields under mild reaction conditions. Moreover, the *N*-allyl-1,2,3-triazole products could be readily reduced to the *N*-alkyl counterparts. Further studies will be reported in due course.

Experimental Section

General Procedure for the Preparation of Compounds 4

Tosyl azide **2a** (0.2 mmol, 1.0 equiv.) was added into a solution of 1,6-allenynes **1** (0.2 mmol, 1.0 equiv.) and CuTc (10 mol%) in toluene (2.0 mL). The mixture was stirred at room temperature for 12 h. Then, H_2O (0.2 mmol, 1.0 equiv.) and TsOH (100 mol%) were successively added at room temperature. After the reaction was completed (monitored by TLC), the resulting mixture was quenched with saturated $NaHCO_3$(aq) and extracted with EtOAc. The combined organic phase was concentrated under reduced pressure and the residue was purified by flash chromatography (silica gel, hexanes/EtOAc/DCM as eluent) to afford the corresponding products **4**.

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[16] CCDC 2076608 (**4a**) contain the crystallographic data for this paper. The data can be obtained free of charge from The Cambridge Crystallographic Data Centre via www.ccdc.cam.ac.uk/data request/cif.

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[18] The moderate yield might be due to two main reasons: several uncharacterized side reactions and the adsorption with silica gel during purification.

[19] The starting material **3b′** was recovered almost quantitatively, and no desired product or desulfonation product was detected.

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